

No Show Paradox and the Golden Number in Generalized Condorcet Voting Methods

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Abstract This paper extends the analysis of the so called no show paradox to the context of generalized q -Condorcet voting correspondences, in which the q -Condorcet winner is defined as the candidate who wins to any other by a qualified majority q higher than half the number of voters. This paradox occurs when a group of voters is better off by not voting than by voting according to its preferences. We try to progress, for the case of general voting correspondences, in the resolution of an open problem proposed in Holzman (Discrete Appl Math 22:133–141, 1988/1989). He asked for the range of q quota values for which all q -Condorcet voting rules are subjected to the paradox. This also means to extend a known general result of Moulin (J Econ Theory 45:53–64, 1988) stating that all conventional Condorcet voting rules ($q = \frac{1}{2}$) suffer the paradox. Interestingly, the well-known mathematical constant φ , called the golden number, appears in the main theorem of the paper. This theorem establishes that a result on voting correspondences of Jimeno et al. (Soc Choice Welf 33:343–359, 2009), similar to Moulin result, does not extend to q -Condorcet voting correspondences if q is equal or higher than $\frac{1}{\varphi}$. More specifically, we find, for every q , a particular correspondence that suffers from the paradox if $q < \frac{1}{\varphi}$, but is free if $q \geq \frac{1}{\varphi}$.

Keywords No show paradox · Participation · Golden number · Generalized Condorcet voting correspondences

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1 Introduction

In the analysis of the voting processes in which a qualified majority is required, an important concept is that of q -Condorcet candidate. This concept is similar to the one already known as “Condorcet candidate”, who is preferred to any other by a majority of voters. A q -Condorcet candidate is preferred to any other by a fraction higher than q of the set of voters, being q a real number in the interval $[\frac{1}{2}, 1)$.

Voting methods consistent with the Condorcet criterion elect the Condorcet candidate whenever there is one. Similarly, we call q -Condorcet methods those which choose the q -Condorcet candidate whenever there is one.

A well known difficulty related to Condorcet methods is their failure on Participation, which can be understood as the requirement that the act of voting does not harm the voter or, in other words, that no voter may manipulate the vote simply by abstaining. This failure of Condorcet methods, called no show paradox, is relevant in the longstanding (and still alive) academic battle between Borda and Condorcet, because the positional methods, and particularly the Borda method, are free from this paradox. See for example, [Nurmi \(2002\)](#), [Risse \(2005\)](#) and [Saari \(2006\)](#) for an exposition of this controversy.

1.1 Related Literature

The basic reference is [Moulin \(1988\)](#), who shows that all Condorcet voting rules, also called resolute Condorcet voting correspondences (which elect a single winner in every voting situation) fail on participation, i.e., there are voting situations where a group of voters is better off by not voting.

Another important reference is [Holzman \(1988/1989\)](#), who shows that the Moulin result is partially extended to a special q -Condorcet voting rules family. Even though this family has been defined from a stronger concept than in this context, [Holzman \(1988/1989\)](#) suggests an open problem, viz. how to extend this result to the general case of q -Condorcet voting rules.

In [Pérez \(2001\)](#) and [Jimeno et al. \(2009\)](#), participation property is adapted to the voting correspondences context (a set of winners, not necessarily a singleton, is chosen in each voting situation), and they show that the negative result of Moulin is still satisfied. In particular, [Jimeno et al. \(2009\)](#) demonstrated that all Condorcet voting correspondences violate the property we call here optimistic participation. That means that no optimistic voter (who thinks that his favourite candidate among the winners will be the only one ultimately chosen) will gain by abstaining. This is the weakest participation property adapted to this new context.

It is worth noting that, since for any candidate the condition of being q -Condorcet (for $q > \frac{1}{2}$) is stronger than being plain Condorcet [or $(\frac{1}{2})$ -Condorcet], for any voting method the consistency property with the q -Condorcet principle, which means choosing such a q -Condorcet candidate if it exists, is weaker than the conventional property. Therefore, the family of methods, called q -Condorcet here, is broader than those called Condorcet methods. In other words, all Condorcet methods are q -Condorcet methods. Nevertheless, in general, q -Condorcet methods are not Condorcet. For exam-

ple, Dodgson method is q -Condorcet, for all $q \geq \frac{1}{2}$, while q -Dodgson (which elects the candidate who requires less favourable exchanges to become the q -Condorcet one) is not a $(\frac{1}{2})$ -Condorcet method, when $q > \frac{1}{2}$.

Therefore, it makes sense to ask whether a result like [Moulin \(1988\)](#) or more precisely like [Jimeno et al. \(2009\)](#), who state that all Condorcet voting methods violate the optimistic participation property, is extended to q -Condorcet methods.

In this paper we explore for what q values this impossibility result is still available. Our main result says that such impossibility disappears for q values equal or above the inverse of the golden number. Specifically, we found a family of voting correspondences we have called the All Not q -Blocked VC which fulfils the optimistic participation property if and only if $q \geq \frac{1}{\varphi}$.

Section 2 presents the basic terminology. Section 3 reviews some known results concerning no show paradox. Section 4 proposes a new family of voting correspondences, already denoted as All Not q -Blocked VC, and establishes results about the possibility or impossibility for this new family to suffer from this paradox, related with the inverse of golden number. Section 5 concludes the paper.

2 Basic Terminology

Let $X = \{x_1, x_2, \dots, x_n, \dots\}$ the set of potential candidates and $V = \{1, 2, \dots, m, \dots\}$ the set of potential voters. Given any finite set of candidates $X = \{x_1, x_2, \dots, x_n\}$, with $n \geq 2$, and any finite set of voters $V = \{1, 2, \dots, m\}$, with $m \geq 1$, preferences of any voter are supposed to be a linear (strict and complete) order L over X . We say $L = x > y > z > t > \dots$, or $L = xyzt\dots$, to denote the preference linear order in which x is the most preferred candidate, y is the second one, and so on, and aLb means that a is preferred to b in L . We call a **situation** any pair (X, p) , where p is a preference profile over X from V , that is to say, an m -tuple of linear orders over X , each one meaning the preferences of a voter over X . By $x >_i y$ we mean that voter i prefers x to y . We call **unanimous** to any situation (X, p) , where all voters in profile p have exactly the same preferences over candidates in X .

Let us call **voting correspondence** (from now on **VC**) any function f which maps any situation (X, p) to a non-empty subset of X , $f(X, p)$. The elements of $f(X, p)$ are the chosen candidates (the winners) over X from the preference profile p . When $f(X, p)$ is required to have only one element for every situation $f(X, p)$, f is said to be a **voting function**, also called voting rule or resolute voting correspondence. Since we will consider only anonymous VCs (all voters are equally considered), a preference profile over X from V can also be described by specifying how many of the m voters from V sustain any of the possible preference orders on X .

Given any X , any two disjoint sets of voters V_1 and V_2 , and any two preference profiles p_1 and p_2 over X from, respectively, V_1 and V_2 , we call $p_1 + p_2$ the new profile over X originated from $V_1 \cup V_2$.

Let (X, p) be any situation with n candidates and m voters. Given any two different candidates x, y from X , $p(x, y)$ is the number of voters in p which strictly prefer x to y . It is obvious that $p(x, y) + p(y, x) = m$. The square $n \times n$ matrix M_p , whose entries are $p(x, y)$ (and without entries in the main diagonal), will be called the **comparison matrix** for (X, p) .

Candidate x is said to **beat** y , if and only if $p(x, y) > p(y, x)$. If, given a situation (X, p) , x beats any other candidate, then x is called the **Condorcet** candidate for this situation. A VC f is called **Condorcet consistent**, or simply **Condorcet**, if it satisfies the **Condorcet criterion**, requiring for every situation (X, p) that, if there is a Condorcet candidate, it will be the only winner.

Given a situation (X, p) , we say that a subset of candidates, $G = \{x_1, x_2, \dots, x_k\} \subset X$, is a **set of clones** in (X, p) if, for all $i \in V$, $x_j >_i z$ implies $x_l >_i z$, for all $x_j, x_l \in G$ and for every $z \in X - G$. That is, a set of clones G is a set of candidates that are identical in the preference of each voter when are compared with respect to any other candidate in $X - G$.

Given a set of clones $G = \{x_1, x_2, \dots, x_k\}$, and a number h of voters (h divisible by k), we define a **balanced cycle** $(x_1, x_2, \dots, x_k)_{k,h}$ of these candidates and voters as the following profile: $\{\frac{h}{k}$ voters: $x_1 x_2 \dots x_{k-1} x_k$; $\frac{h}{k}$ voters: $x_2 x_3 \dots x_k x_1$; $\frac{h}{k}$ voters: $x_3 x_4 \dots x_1 x_2$; $\dots$; $\frac{h}{k}$ voters: $x_k x_1 \dots x_{k-2} x_{k-1}\}$. Its comparison matrix is:

	x_1	x_2	x_3	$\dots$	x_{k-1}	x_k
x_1		$\frac{(k-1)h}{k}$	$\frac{(k-2)h}{k}$	$\dots$	$\frac{2h}{k}$	$\frac{h}{k}$
x_2	$\frac{h}{k}$		$\frac{(k-1)h}{k}$	$\dots$	$\frac{3h}{k}$	$\frac{2h}{k}$
x_3	$\frac{2h}{k}$	$\frac{h}{k}$		$\dots$	$\frac{4h}{k}$	$\frac{3h}{k}$
$\dots$	$\dots$	$\dots$	$\dots$		$\dots$	$\dots$
x_{k-1}	$\frac{(k-2)h}{k}$	$\frac{(k-3)h}{k}$	$\frac{(k-4)h}{k}$	$\dots$		$\frac{(k-1)h}{k}$
x_k	$\frac{(k-1)h}{k}$	$\frac{(k-2)h}{k}$	$\frac{(k-3)h}{k}$	$\dots$	$\frac{h}{k}$	

where the minimum entry of each row is $\frac{h}{k}$. Note that a balanced cycle is a $\frac{h}{k}$ replication of the Saari's ranking wheel of the set of candidates (Saari 2011).

3 No Show Paradox in Condorcet Voting Methods

The no show paradox appears in the literature as a flaw of voting methods that fail to satisfy a property of participation, i.e., where voters are better off by voting than by abstaining. Although this paradox affects to different types of voting methods, it has been shown, as we can see below, that it is a special flaw in Condorcet methods.

In the context of voting rules (resolute voting methods), Moulin (1988) defines the **participation property** as follows:

Definition 1 A Voting function f satisfies the **participation** property if for any given pair of situations (X, p) and (X, v) , where profile v has only one voter, $[f(X, p) = \{x\}$ and x is preferred to y in $v]$ implies $f(X, p + v) \neq \{y\}$.

In words, when f satisfies participation, if x is the winner for a situation and a new voter who prefers x to y is added, candidate y will not become the winner. Thus, the new voter could not do better abstaining, because submitting his ballot would never result in the election of a less preferred candidate. In this sense, failing to satisfy

participation means that the no show paradox sets in because a potential voter could do better abstaining.

The incompatibility of the above property with the Condorcet criterion is shown in [Moulin \(1988\)](#), who established that no Condorcet voting function satisfies the participation property.

This result has been extended in several directions. Within the context of voting functions [Holzman \(1988/1989\)](#) obtains it by the definition, for each situation (X, p) and $q \in [\frac{1}{2}, 1)$, of a set called q -Core. This set contains every candidate x such that no candidate y beats him by a fraction q of voters. And he requires that the voting function selects a candidate belonging to q -Core as winner, provided this set is not empty.

Then, [Holzman](#) shows that all voting functions obeying the above condition satisfy Participation if and only if $q \geq \frac{n-1}{n}$ or $n \leq 3$ where n is the number of candidates.

As [Nurmi \(2004\)](#) indicates, in this case, provided that the number of candidates is large enough, there is no way to avoid the paradox unless we demand some quite high qualified majority.

Notice that $q = \frac{1}{2}$ leads to the Moulin result.

In the context of voting correspondences (irresolute voting methods), there are different ways to define participation depending on the preferences of voters over subsets of candidates, as we can see in [Pérez \(1995, 2001\)](#) and [Jimeno et al. \(2009\)](#).

For our purposes, more interesting is a property called **optimistic participation** which is defined, among others, in [Jimeno et al. \(2009\)](#) and [Pérez et al. \(2012\)](#) as **immunity to manipulation by abstention in the sense of Optimistic Order**.

Definition 2 A VC f satisfies the **optimistic participation (OptP)** property if for any given pair of situations (X, p) and (X, v) , where situation (X, v) is unanimous, there is $y \in f(X, p + v)$ such that no $x \in f(X, p)$ is strictly preferred to y in v .

In words, *OptP* means that when some voters with equal preferences are added to the initial situation, we can be sure that there will be a winner candidate who is not worse for those voters than the best of the previous winners. If this property is satisfied, we can be sure that no optimistic voter (a voter who thinks that his favourite candidate among the winners will be the only one ultimately chosen) will gain by abstaining.

Failing to satisfy this property means the appearance of the “optimistic version of no show paradox” in voting correspondences. And the failure of this property affects to all Condorcet voting correspondences as is shown in [Jimeno et al. \(2009\)](#).

Theorem 1 ([Jimeno et al. 2009](#)) *No Condorcet VC satisfies the OptP property.*

4 New Results in q -Condorcet Voting Correspondences and the Golden Number

In this section we will explore the possibility of extending known results about the incompatibility between Condorcet and participation-like properties to voting correspondences based on a generalization of Condorcet criterion that we will call q -Condorcet voting methods.

Given a situation (X, p) , we say that candidate x “defeats or wins by a strict q -majority” to another candidate y , if $p(x, y) > qm$, where $q \in [\frac{1}{2}, 1)$ and m is the

number of voters. Candidate x is a q -winner candidate if x defeats any other candidate by a strict q -majority. When $q \geq \frac{1}{2}$ a q -winner candidate will be called a **q -Condorcet winner** or just q -Condorcet. In the particular case $q = \frac{1}{2}$ the q -Condorcet winner is just the Condorcet candidate in the conventional sense defined above, while when q satisfies $\frac{m-1}{m} \leq q < 1$ the q -Condorcet winner is a candidate who Pareto dominates any other candidate.

Definition 3 Given a fixed number q in the interval $[\frac{1}{2}, 1)$, we say that a voting correspondence f is **q -Condorcet**, or consistent with the **q -generalized Condorcet criterion**, when for all situation (X, p) , if x is a q -Condorcet candidate, then $f(X, p) = \{x\}$.

In words, f is q -Condorcet if whenever a q -Condorcet candidate exists he will be the only chosen. Note that this definition poses no a priori restriction on the winner set when such a q -winner candidate does not exist. Thus, to be a q -Condorcet method is less demanding as the value of q increases.

We have restricted the Condorcet winner concept, thus weakening the Condorcet criterion concept. It seems clear that, given $q_1, q_2 \in [\frac{1}{2}, 1)$, such that $q_1 \leq q_2$, every q_1 -Condorcet correspondence is also a q_2 -Condorcet correspondence. In particular, given $q \in [\frac{1}{2}, 1)$, if the f correspondence is Condorcet then it will also be q -Condorcet.

Let us clarify the previous point with the following example: If f is a $(\frac{2}{3})$ -Condorcet method, it will select the candidate defeating any other candidate by a strict $\frac{2}{3}$ -majority, as the sole winner. But if that candidate does not exist, no demand is made on the winner set, and it can turn out that a Condorcet winner who defeats almost all the others by a strict $\frac{2}{3}$ -majority is not chosen.

Once we have defined the family of q -Condorcet voting correspondences we may define a special kind of candidates that we call q -blocked candidates:

Definition 4 Given a situation (X, p) , we say that candidate x is **q -blocked** by candidate z in (X, p) if there exists a potential voter $v: x \dots z \dots$ such that z is the q -Condorcet candidate in the new situation $(X, p + hv)$, where hv represents a profile of h identical voters with preference v .

A q -blocked candidate x is a candidate so inferior to candidate z blocking him that, as new voters are added, z can become the q -Condorcet winner ahead of x even in cases in which x is the favourite of these voters. By Definition 4, when a situation has a q -Condorcet candidate, all other candidates are blocked by him.

In Lemma 1, which we do not prove, is shown that electing a q -blocked candidate by any q -Condorcet VC leads to the no show paradox.

Lemma 1 Let $q \in [\frac{1}{2}, 1)$, f_q any q -Condorcet VC and (X, p) a situation in which x is q -blocked by z . If f_q satisfies OptP then $x \notin f_q(X, p)$.

The following lemma characterizes q -blocked candidates using only the q -comparison matrix information of the initial situation.

Lemma 2 Let $q \in [\frac{1}{2}, 1)$. Given any situation (X, p) with m voters and any two candidates x and z , x is q -blocked by z in (X, p) if and only if

$$\text{Min}_{y \in X} p(z, y) > \frac{2q-1}{q}m + \frac{1-q}{q}p(x, z) \quad (1)$$

Proof Consider the case when z is the q -Condorcet candidate in (X, p) . In this case any x is q -blocked by z and the inequality (1) holds obviously.

Suppose now that z is not a q -Condorcet candidate. Let (X, p) be any situation with m voters in the profile p and $X = \{x_1, x_2, \dots, x_n\}$ where $x_1 = x$ and $x_2 = z$, in which

$$c > \frac{2q-1}{q}m + \frac{1-q}{q}a \quad (2)$$

where $c = \text{Min}_{y \in X} p(z, y)$ and $a = p(x, z)$.

Since

$$\frac{2q-1}{q}m + \frac{1-q}{q}a = \frac{2q-1}{q}m + \frac{1-q}{q}(m + a - m) = m - \frac{1-q}{q}(m - a),$$

inequality (2) is equivalent to:

$$m - a > \frac{q}{1-q}(m - c) \quad (3)$$

And since

$$c > \frac{2q-1}{q}m + \frac{1-q}{q}a$$

means $qc > 2qm - m + a - q$, the inequality (2) is also equivalent to

$$m - a > q(2m - c - a) \quad (4)$$

Let $h = p(z, x) - \text{Min}_{y \in X} p(z, y) = m - a - c$. Since z is not a q -Condorcet candidate, $c \leq qm$. By (3),

$$h = m - a - c > \frac{q}{1-q}(m - c) - c = m \frac{q}{1-q} - c \frac{1}{1-q} \geq 0.$$

Hence $h > 0$.

Let us add $h = m - a - c$ new voters to the situation (X, p) , all with identical preference order $v: xz \dots$. In the new situation (X, p') , we have the following results:

- (a) Condition (4) is equivalent to $p'(z, x) > q(2m - c - a) = q(m + h)$. Therefore, if we compare candidates z and x ,

$$p'(z, x) = p(z, x) = m - a > q(m + h).$$

- (b) Again, condition (4) is equivalent to $p'(z, t) > q(2m - c - a) = q(m + h)$. Thus, if now we compare candidates z and t , for all $t \neq x, z$, $p'(z, t) = p(z, t) + h \geq m - a > q(m + h)$.

Hence, z is a q -Condorcet candidate in the new situation (X, p') .

Therefore, in any case x is q -blocked by z if and only if inequality (1) holds. $\square$

For $q = \frac{1}{2}$, condition (1) is reduced to $\text{Min}_{y \in X} p(z, y) > p(x, z)$, so that Lemma 2 is reduced to Lemma 1 of Jimeno et al. (2009) for Condorcet voting correspondences, similar to a claim in Moulin (1988) for Condorcet voting rules.

A q -blocked candidate has a big weakness against the candidate who blocks him. This turns him in a weak candidate against other candidates in all Condorcet and q -Condorcet methods based on pairwise comparison. The following lemma shows that the condition of not being blocked by another concrete candidate has certain stability against the introduction of new voters for whom this other candidate is least preferred:

Lemma 3 *Let $q \in [\frac{1}{2}, 1)$. Given any situation (X, p) with m voters, if x is not q -blocked by z in (X, p) , then x will not be q -blocked by z in a new situation $(X, p + hv)$, where h new identical voters have been added with preferences $v: \dots x \dots z \dots$.*

Proof Let (X, p) and (X, p') two situations with m and $m + h$ voters, respectively, where $p' = p + hv$ and h are identical votes with preferences $v: \dots x \dots z \dots$. Then, $p'(x, z) = h + p(x, z)$ and

$$\text{Min}_{y \in X} p'(z, y) \leq h + \text{Min}_{y \in X} p(z, y).$$

Therefore, if x is not q -blocked by z in (X, p) , then by Lemma 2,

$$\text{Min}_{y \in X} p(z, y) \leq \frac{2q - 1}{q}m + \frac{1 - q}{q}p(x, z).$$

So that,

$$\begin{aligned} \text{Min}_{y \in X} p'(z, y) &\leq h + \text{Min}_{y \in X} p(z, y) \leq \frac{2q - 1}{q}m + \frac{1 - q}{q}p(x, z) + h \\ &= \frac{2q - 1}{q}(m + h) + \frac{1 - q}{q}p'(x, z). \end{aligned}$$

Then, by Lemma 2, x is not q -blocked by z in (X, p') . $\square$

Lemma 3 tells us that if one candidate x is q -blocked by another candidate z , and there are voters with preferences $v: \dots x \dots z \dots$, in p , this candidate is also q -blocked in a situation obtained deleting some or all those voters.

Since q -blocked candidates are weak against other candidates, we can consider, as an alternative selection method, a voting correspondence that does not choose any of them, such as the following voting correspondence called *All Not q -Blocked VC*:

Definition 5 Given a fixed number q in the interval $[\frac{1}{2}, 1)$, we call **All Not q -Blocked Correspondence** (f_{q-ANB}) to the voting correspondence that selects as winners for any situation all not q -blocked candidates in that situation.

It is easy to show that for each value of q in the range $[\frac{1}{2}, 1)$, the f_{q-ANB} correspondence is q -Condorcet. Indeed, if in a (X, p) situation, candidate z is a q -Condorcet candidate, this candidate blocks all others, thereby becoming the only winner. More formally, $p(z, y) > qm, \forall y \in X$ implies $\text{Min}_{y \in X} p(z, y) > \frac{2q-1}{q}m + \frac{1-q}{q}p(x, z)$ because $p(x, z) < (1 - q)m$.

The following lemma relates the failure of the optimistic participation property by this voting correspondence with the existence of concatenations of q -blocked candidates, when the last blocking candidate is not a q -Condorcet candidate.

Lemma 4 Given a fixed number q in the interval $[\frac{1}{2}, 1)$, let (X, p) and $(X, p + hv)$ be two situations showing a failure of f_{q-ANB} to satisfy *OptP*. Then there exists a r -concatenation of q -blocked candidates $x_1 x_2 \dots x_r$ in $(X, p + hv)$, with $r \geq 2$, i.e., candidate x_1 is q -blocked by x_2 , x_2 is q -blocked by x_3 , ..., x_{r-1} is q -blocked by x_r , such that x_r is not a q -Condorcet in $(X, p + hv)$.

Proof Given a fixed number q in the interval $[\frac{1}{2}, 1)$, let (X, p) and $(X, p + hv)$ two situations, with preference of voter $v: \dots x \dots z \dots$, such that x is his favourite winner in $f_{q-ANB}(X, p)$ and z is his favourite winner in $f_{q-ANB}(X, p + hv)$. So that x is q -blocked by a candidate x_1 in $(X, p + hv)$. And then $x_1 >_v x$, because otherwise, by Lemma 3, x would have been also q -blocked by candidate x_1 in (X, p) . Therefore, since $x_1 \notin f_{q-ANB}(X, p)$, there must be another candidate x_2 , such that x_1 is q -blocked by x_2 in $(X, p + hv)$ and, at the same time, x is q -blocked by x_1 in $(X, p + hv)$. This concatenation of q -blockings would lead eventually to a candidate x_r such that $x >_v x_r$.

And candidate x_r could not be q -Condorcet, otherwise x_r would q -block all other candidates (including x) in $(X, p + hv)$, implying, by Lemma 3, that x_r would q -block x in p , what is in contradiction with $x \in f_{q-ANB}(X, p)$.

Thus, the failure to satisfy *OptP* lead to a concatenation of q -blockings, and the absence of such concatenations implies that f_{q-ANB} satisfies *OptP*. $\square$

Once we have established the relationship between the existence of concatenations without q -Condorcet candidate and the failure of $f_{q-ANB}(X, p)$ on optimistic participation, we are interested to find out if there is a q quota value from which these concatenations no longer exist. The answer, as indicated in Lemma 5, is the inverse of the golden number φ .

Let us recall some facts about the famous golden number φ . It is the irrational number $\frac{1+\sqrt{5}}{2}$, which is the only positive solution of the equation $q - 1 = \frac{1}{q}$. Its decimal expansion is 1.61803... and can also be computed as the limit of the rational sequence $\frac{F(n+1)}{F(n)}$, where $F(n)$ is the n th term of the Fibonacci sequence. Finally, by definition, the inverse of the golden number is $\varphi - 1 = 0.61803 \dots$

Lemma 5 If $q \geq \frac{1}{\varphi}$ there is no situation (X, p) with 2-concatenation xyz such that z is not a q -Condorcet candidate.

Proof Given a fixed number q in the interval $[\frac{1}{2}, 1)$, let (X, p) a situation with a 2-concatenation xyz , i.e., where x is q -blocked by y and y is q -blocked by z . Then by Lemma 2:

$$\begin{aligned} \text{Min}_{u \in X} p(y, u) &> \frac{2q-1}{q}m + \frac{1-q}{q}p(x, y) \quad \text{and} \\ \text{Min}_{v \in X} p(z, v) &> \frac{2q-1}{q}m + \frac{1-q}{q}p(y, z). \end{aligned}$$

Hence, since $p(x, y) \geq 0$ and $p(y, z) \geq \text{Min}_{u \in X} p(y, u)$, $\text{Min}_{v \in X} p(z, v) > \frac{2q-1}{q}m + \frac{1-q}{q}\text{Min}_{u \in X} p(y, u) \geq \frac{2q-1}{q^2}m$. Thus, $\frac{\text{Min}_{v \in X} p(z, v)}{m} > \frac{2q-1}{q^2}$. And because z is not q -Condorcet candidate, i.e., $\frac{\text{Min}_{v \in X} p(z, v)}{m} \leq q$, we obtain $q > \frac{2q-1}{q^2}$ or equivalently $q^3 - 2q + 1 > 0$. Finally, since $q^3 - 2q + 1 = (q-1)(q^2 + q + 1)$, we can say equivalently that $q^2 + q - 1 < 0$.

Therefore, if xyz is a 2-concatenation where z is not a q -Condorcet candidate, then $q^2 + q - 1 < 0$. And this inequality is satisfied if and only if $\frac{-1-\sqrt{5}}{2} = -\varphi < q < \frac{-1+\sqrt{5}}{2} = \frac{1}{\varphi}$.

Concluding, if the described 2-concatenation exists then $q < \frac{1}{\varphi}$. Equivalently, if $q \geq \frac{1}{\varphi}$ then there is no such 2-concatenation. Obviously, this implies also that there is no r -concatenation with $r > 2$, because in such a case the q -Condorcet candidate would be q -blocked. $\square$

From these lemmas it will be easy to prove the first part of Theorem 2 (if), establishing that when q is equal or higher than $\frac{1}{\varphi}$, the VC f_{q-ANB} satisfies optimistic participation. The second part (only if) states that for any q in the interval $[\frac{1}{2}, \frac{1}{\varphi})$, the VC f_{q-ANB} fails optimistic participation.

Theorem 2 Given any q in $[\frac{1}{2}, 1)$, the VC f_{q-ANB} satisfies *OptP* if and only if $q \geq \frac{1}{\varphi}$.

Proof (if) Let $q \geq \frac{1}{\varphi}$. By Lemma 5 there is no situation (X, p) with a 2-concatenation xyz such that z is not a q -Condorcet candidate. Therefore, by Lemma 4, f_{q-ANB} satisfies *OptP*.

(only if) For reasons of computational simplicity we will allow the numbers of the different groups of voters to be non-integer and even non-rational, in a similar way than Holzman (1988/1989) in the proof of his principal result.

Hence we will consider situations with fractional votes and, consequently, fractional entries in M_P . Thus, if $p(x, y) + p(y, x) = 1$, $p(x, y)$ means the proportion of voters strictly preferring x to y . We can do that because the concepts and properties used in this paper are invariants under multiplication of every entry in M_P by any real positive factor $\gamma > 0$.

Let $q \in [\frac{1}{2}, \frac{1}{\varphi})$. For every q , we will find a situation in which f_{q-ANB} fails *OptP*. Let the total number of voters be $m = 1$, and the set of candidates be $X = \{x, y, z; t_1, t_2, t_3\}$, where $[t] = \{t_1, t_2, t_3\}$ is a group of clones. Inside this

group of clones, the voter preferences are those of a balanced cycle, while the preferences between x , y , z , and among x , y , z , and $[t]$, are: $\{\frac{1}{\varphi}$ voters: $[t]zyx$; $1 - \frac{1}{\varphi} - \varepsilon$ voters: $zyx[t]$; ε voters: $xzy[t]$, where $0 < \varepsilon < \frac{1}{\varphi}\}$. The corresponding abbreviated comparison matrix is:

$M_{P(x,y,z,[t])}$	x	y	z	$[t]$
x		ε	ε	$1 - \frac{1}{\varphi}$
y	$1 - \varepsilon$		0	$1 - \frac{1}{\varphi}$
z	$1 - \varepsilon$	1		$1 - \frac{1}{\varphi}$
$[t]$	$\frac{1}{\varphi}$	$\frac{1}{\varphi}$	$\frac{1}{\varphi}$	$M_{P([t])}$

and the corresponding additional comparison matrix $M_{P([t])}$ is:

$M_{P([t])}$	t_1	t_2	t_3
t_1		$\frac{2}{3}$	$\frac{1}{3}$
t_2	$\frac{1}{3}$		$\frac{2}{3}$
t_3	$\frac{2}{3}$	$\frac{1}{3}$	

Observe that the minimal entry of the row of every t_i is $\frac{1}{3}$.

Let us also consider the situation (X, p') obtained adding, to the above situation, α new voters with preferences: $yxzt_1t_2t_3$. The corresponding abbreviated comparison matrix is now:

$M_{P'(x,y,z,[t])}$	x	y	z	$[t]$
x		ε	$\varepsilon + \alpha$	$1 - \frac{1}{\varphi} + \alpha$
y	$1 - \varepsilon + \alpha$		α	$1 - \frac{1}{\varphi} + \alpha$
z	$1 - \varepsilon$	1		$1 - \frac{1}{\varphi} + \alpha$
$[t]$	$\frac{1}{\varphi}$	$\frac{1}{\varphi}$	$\frac{1}{\varphi}$	$M_{P'([t])}$

Now, the additional comparison matrix $M_{P'([t])}$ lacks any relevance. We are going to prove now that, given any fixed q in $[\frac{1}{2}, 1)$, there exist ε and α , where $0 < \alpha$ and $0 < \varepsilon < \frac{1}{\varphi}$, such that the following facts are true:

- (1) Candidate x is not q -blocked by any other candidate in the initial situation (X, p) .
- (2) Candidate x is q -blocked by y in the posterior situation (X, p') .
- (3) Candidate y is q -blocked by z in the posterior situation (X, p') .

In order to apply Lemma 2, note that the minimal entry in the row of candidate z in M_P is $1 - \frac{1}{\varphi}$. Moreover, since the minimal entry in the row of every candidate in $[t]$ is

$\frac{1}{3}$, less than $p(x, [t]) = 1 - \frac{1}{\varphi}$, no candidate in $[t]$ q -blocks x in (X, p) . Note also that α is the minimal entry of the y row in $M_{p'}$ and that, provided that $\varepsilon \geq \frac{1}{\varphi} - \alpha$, $1 - \varepsilon$ is the minimal entry of the z row in $M_{p'}$.

By Lemma 2, the following are sufficient conditions for these facts to be true:

$$1 - \frac{1}{\varphi} \leq \frac{2q-1}{q} + \frac{1-q}{q}\varepsilon \quad (5)$$

$$\alpha > \frac{2q-1}{q}(1+\alpha) + \frac{1-q}{q}\varepsilon \quad (6)$$

$$1 - \varepsilon > \frac{2q-1}{q}(1+\alpha) + \frac{1-q}{q}\alpha \quad (7a)$$

$$\varepsilon \geq \frac{1}{\varphi} - \alpha \quad (7b)$$

Conditions (5) and (6) imply, respectively, that $x \in f_{q-ANB}(X, p)$ and $x \notin f_{q-ANB}(X, p')$. Conditions (7a) and (7b) imply that $y \notin f_{q-ANB}(X, p')$. When these four conditions hold, a no show paradox occurs (at least a theoretical one, given the possible non-integer nature of the numbers), because when the α voters abstain, x is elected, but when they vote only less preferred candidates are elected.

Clearing the unknown ε in conditions (5), (6), (7a) and (7b), we obtain:

$$\varepsilon \geq \frac{q\left(1 - \frac{1}{\varphi}\right) - 2q + 1}{1 - q} \quad (5^*)$$

$$\varepsilon < \alpha - \frac{2q-1}{1-q} \quad (6^*)$$

$$\varepsilon < \frac{1-q}{q} - \alpha \quad (7a^*)$$

$$\varepsilon \geq \frac{1}{\varphi} - \alpha \quad (7b^*)$$

In the $\alpha - \varepsilon$ plane, the border of (5*) is a horizontal line, while the borders of (6*) and (7a*) are straight lines with slopes 1 and -1 , and intersects $-\frac{2q-1}{1-q}$ and $\frac{1-q}{q}$, respectively. If the first three conditions hold, these three border lines will form an isosceles triangle RTS with a horizontal base and nonempty interior, the location of whose vertices depends on q . Let us prove that, for any given q in $\left[\frac{1}{2}, \frac{1}{\varphi}\right)$, these three conditions are compatible. The intersection points between these three border lines are

$$(5^*) \text{ and } (6^*): R \left(\alpha = \frac{q\left(1 - \frac{1}{\varphi}\right)}{1-q}, \varepsilon = \frac{q\left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q} \right),$$

$$(5^*) \text{ and } (7a^*): S \left(\alpha = \frac{1-q}{q} - \frac{q \left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q}, \right. \\ \left. \varepsilon = \frac{q \left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q} \right), \text{ and} \\ (6^*) \text{ and } (7a^*): T \left(\alpha = \frac{1-q}{2q} + \frac{(2q-1)}{2(1-q)}, \varepsilon = \frac{1-q}{2q} - \frac{(2q-1)}{2(1-q)} \right)$$

Let us prove now that the second coordinate of point T , $\varepsilon_T = \frac{1-q}{2q} - \frac{(2q-1)}{2(1-q)}$, is strictly higher than that of the other two points, $\varepsilon_R = \varepsilon_S = \frac{q \left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q}$:

$$\varepsilon_T > \varepsilon_R \Leftrightarrow \frac{1-q}{2q} - \frac{(2q-1)}{2(1-q)} > \frac{q \left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q} \Leftrightarrow \frac{3q-1-q^2}{2q^2} < \frac{1}{\varphi} \quad (8)$$

To see if (8) holds, let us analyze the function $h(q) = \frac{3q-1-q^2}{2q^2}$. This function is analytical in the interval $(0, +\infty)$ and has only one critical point at $q = \frac{2}{3}$, reaching a maximum in it. On the other hand, $h(1 - \frac{1}{\varphi}) = 0$, $h(\frac{1}{2}) = \frac{1}{2}$, $h(q)$ is strictly increasing in the interval $[\frac{1}{2}, \frac{1}{\varphi})$, and, finally, $h(\frac{1}{\varphi}) = \frac{1}{\varphi}$. Therefore, (8) holds in $[\frac{1}{2}, \frac{1}{\varphi})$.

Thus, we can be sure that the triangle RTS has a nonempty interior.

Finally, let us prove that condition (7b*) is redundant with the other three, showing that its border runs below point R . The border of (7b*) is a straight line with slope -1 and intersect $\frac{1}{\varphi}$. Computing for this line the value ε corresponding to the value of α in R , $\alpha = \frac{q(1-\frac{1}{\varphi})}{1-q}$, we obtain $\varepsilon = \frac{1}{\varphi} - \frac{q(1-\frac{1}{\varphi})}{1-q}$. Comparing this value with that of ε in point R , $\varepsilon = \frac{q(1-\frac{1}{\varphi}) - (2q-1)}{1-q}$, we have:

$$\frac{1}{\varphi} - \frac{q \left(1 - \frac{1}{\varphi}\right)}{1-q} < \frac{q \left(1 - \frac{1}{\varphi}\right) - (2q-1)}{1-q} \Leftrightarrow q < \varphi - 1 = \frac{1}{\varphi}$$

Thus, since $q < \frac{1}{\varphi}$ by hypothesis, (7b*) holds if (5*), (6*) and (7a*) hold.

Therefore, for every $q < \frac{1}{\varphi}$, there exist two real numbers α and ε , both dependent on q , for which the situations described above lead to the paradox. It would be easy to show that, as q approaches to $\frac{1}{\varphi}$, the corresponding triangle shrinks and, at the same time, approaches to the point $Q = (\frac{1}{\varphi}, 0)$.

It is necessary to observe that in the above reasoning, the coordinates of points R , S and T depend, not only on q , but also on φ . Nevertheless, we can say, based on a continuity argument, that if we replace φ by a rational number δ_q sufficiently near

to φ , the new triangle $R_q S_q T_q$ keep having a nonempty interior, only dependent on q , and condition (7b*) keep being redundant with the other three.

Furthermore, because of the known property of the rational numbers of being dense in the set of real numbers, we can conclude that there exist, for every q in $(\frac{1}{2}, \frac{1}{\varphi})$, three rational numbers α_q , ε_q and δ_q , all dependent on q , such that a situation as the described above, but with rational entries, exist. If we multiply all the numbers in this situation by an appropriate natural number, we will obtain a “real practical situation” where the size of each group of voters is a natural (though perhaps high) number of voters. Such situation exhibits the failing of *OptP* by the correspondence f_{q-ANB} , for the corresponding q . $\square$

We have shown that the set of voting correspondences f_{q-ANB} with $q \geq \frac{1}{\varphi}$, satisfy optimistic participation. Therefore this contradicts, for $q \geq \frac{1}{\varphi}$, the extension of the overall impossibility result described in Theorem 1.

However, it is still open the possibility that other q -Condorcet voting correspondences are free from the paradox, even for q values less than $\frac{1}{\varphi}$.

In the second part of Theorem 2 we have needed to construct a series of feasible regions (triangles) with nonempty interior, one for each value of q , and each one with infinite instances of the paradox. Perhaps there is a more simplified and practical way of showing an easy instance for every q , but we could not find it. However, it is easy to find simple examples for particular values of q , when these values are not too close to $\frac{1}{\varphi}$.

We will not repeat the well-known case of $q = \frac{1}{2}$. To clarify the results obtained, the following example, with only 17 voters and 5 candidates, illustrates how f_{q-ANB} fails to satisfy optimist participation, when q is less than $\frac{1}{\varphi}$.

Example Let $q = 0.6$ and let (X, p) a situation with $m = 17$ voters and $n = 5$ candidates, where voters preference profile is $p: \{9 \text{ voters: } t_1 t_2 z y x; 1 \text{ voter: } t_2 t_1 z y x; 6 \text{ voters: } z y x t_2 t_1; 1 \text{ voter: } t_2 t_1 x z y\}$.

As result, the comparison matrix is:

M_P	x	y	z	t_1	t_2
x		1	1	6	6
y	16		0	6	6
z	16	17		6	6
t_1	11	11	11		9
t_2	11	11	11	8	

We can see that:

$$\begin{aligned}
 \text{Min}_{u \in X} p(y, u) &= 0 < \frac{2q-1}{q} 17 + \frac{1-q}{q} p(x, y) \\
 &= \frac{2(0.6)-1}{(0.6)} 17 + \frac{1-(0.6)}{(0.6)} 1 = 6.\hat{3}
 \end{aligned}$$

$$\begin{aligned}
\text{Min}_{u \in X} p(z, u) &= 6 < \frac{2q-1}{q} 17 + \frac{1-q}{q} p(x, z) \\
&= \frac{2(0.6)-1}{(0.6)} 17 + \frac{1-(0.6)}{(0.6)} 1 = 6.\hat{3} \\
\text{Min}_{u \in X} p(t_1, u) &= 9 < \frac{2q-1}{q} 17 + \frac{1-q}{q} p(x, t_1) \\
&= \frac{2(0.6)-1}{(0.6)} 17 + \frac{1-(0.6)}{(0.6)} 6 = 9.\hat{6} \\
\text{Min}_{u \in X} p(t_2, u) &= 8 < \frac{2q-1}{q} 17 + \frac{1-q}{q} p(x, t_2) \\
&= \frac{2(0.6)-1}{(0.6)} 17 + \frac{1-(0.6)}{(0.6)} 6 = 9.\hat{6}
\end{aligned}$$

Therefore, x is not q -blocked by any candidate in (X, p) , i.e. $x \in f_{0.6-ANB}(X, p)$. If we add 10 new voters with preferences $v: yxz t_2 t_1$, the new comparison matrix is:

M_{P+10v}	x	y	z	t_1	t_2
x		1	11	16	16
y	26		10	16	16
z	16	17		16	16
t_1	11	11	11		9
t_2	11	11	11	18	

Now, we can show that:

$$\begin{aligned}
\text{Min}_{u \in X} p'(y, u) &= 10 > \frac{2q-1}{q} 27 + \frac{1-q}{q} p'(x, y) \\
&= \frac{2(0.6)-1}{(0.6)} 27 + \frac{1-(0.6)}{(0.6)} 1 = 9.\hat{6} \\
\text{Min}_{u \in X} p'(z, u) &= 16 > \frac{2q-1}{q} 27 + \frac{1-q}{q} p'(y, z) \\
&= \frac{2(0.6)-1}{(0.6)} 27 + \frac{1-(0.6)}{(0.6)} 10 = 15.\hat{6}
\end{aligned}$$

In the new situation, we see that x is 0.6-blocked by y , and simultaneously y is 0.6-blocked by z . In consequence, neither x nor y belong to $f_{0.6-ANB}(X, p + 10v)$. Therefore, $f_{0.6-ANB}$ fails optimistic participation.

It would be easy to find similar examples for higher values of q . In particular, for $q = 0.61$, 35 voters and 5 candidates are sufficient to show the failure of optimistic participation.

5 Conclusions and Remarks

In this paper we have explored the possibility of extending to all q -Condorcet voting correspondences with q in $[\frac{1}{2}, 1)$, the result already known for $q = \frac{1}{2}$, according to which all of them are subjected to the optimistic no show paradox. This exploration has led us to discover a family of voting correspondences, which we have called All Not q -Blocked VC, denoted f_{q-ANB} , whose relationship with this paradox depends crucially on the comparison between q and the golden number φ .

More specifically, Theorem 2 shows, for every q in $[\frac{1}{2}, 1)$, that f_{q-ANB} suffers from the paradox if $q < \frac{1}{\varphi}$, but is free if $q \geq \frac{1}{\varphi}$.

Although we have used the term q -Condorcet, our approach is quite different from others considered in voting and social choice literature. For example, Holzman (1988/1989), Nurmi (2002, 2004) or Baharad and Nitzan (2003), among others, use the term “generalized Condorcet candidate”, for $q \geq \frac{1}{2}$. But it is worth noting that this is usually done in the sense of weakening the Condorcet candidate concept, i.e., not claiming that this candidate is preferred to any other by a fraction q of voters, but requiring that no other is preferred to him by this majority. In such cases, the corresponding Condorcet voting methods are more demanding, and belong to the conventional Condorcet family.

It is also worth emphasizing three main differences between this work and Moulin (1988) and Holzman (1988/1989). The first one is that they study voting rules, while our object of analysis is voting correspondences. The second one is that they search to identify not only what voting rules suffer the paradox, but how many candidates and voters are required for the paradox to appear. Instead we content with determining which voting correspondences suffer the paradox, even if a large number of candidates and voters may be necessary. The third one is that Holzman (1988/1989) fixes the number of candidates before he defines the participation property. This allows him to show that, given q and given an n number of candidates, certain voting rules fulfil Participation if $q \geq \frac{n-1}{n}$. However, with our definition of optimistic participation, we can claim that they do not fulfil it. Indeed, the Holzman result establishes the existence of situations, with a larger number of candidates, in which the paradox occurs.

On the other side, we can stand out that the well-known golden number also appears in a meaningful way in the Social Choice context.

Given the ubiquity of the golden number and the Fibonacci sequence, it is not surprising to find it in fields closed to our research. We have seen the closest appearance in the field of simple voting games, where Freixas and Kurz (2013) show that the Fibonacci sequence and the golden number play an important role in the enumeration of all non-isomorphic games for same special classes of simple games.

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